BharatTrip Refund Operations: Diagnosis, Architecture & Measurement

Role: AI Operations Associate | **Scope:** Screening Take-Home Task | **Constraint:** Zero Headcount Addition

1. Problem Definition: The Root Cause Behind the Escalation Surge

The Symptom vs. Root Reality: Leadership observed a critical escalation surge over May–June 2026 while overall refund intake volume remained flat (~145 requests/month). The standard organizational reflex is to assume an agent performance slump or understaffing. However, empirical analysis of the operational extract reveals that the escalation crisis is driven by **three decoupled structural failures** across systems, status semantics, and approval queues:

- **Disconnected Operational Silos & Asynchronous Trackers:** Support and Finance maintain separate, non-reconciled spreadsheets. Support agents mark tickets *'Closed'* the moment they respond to a travel agent or forward an email. However, **69 tickets logged by Support never reached Finance**. Travel agents are told their refund is 'done', while Finance has zero record of the request.
- **June Approval Choke-Point:** Finance approvals completely collapsed in June. Prior to June, pending payouts at month-end were zero. In June, **58 payouts stalled in *'Pending Payout'*** (accounting for 47% of June Finance volume). Approver *N. Iyer* ceased sign-offs mid-June without a delegation failover, creating a silent queue freeze.
- **Omnichannel Shadow Leakage & Unlogged Ingestion:** Informal WhatsApp and email cancellation requests (e.g., *RF-1098* and *RF-1099*) were acknowledged verbally without issuing a canonical ticket ID or creating a record in either tracker. When agents chased two weeks later, Support had no reference, driving instant critical escalations.
- **Deduction Transparency & Amount Gaps:** Across matched completed payouts, 53 cases had a gap where requested amount exceeded paid amount. Standard cancellation deductions explain 556 of 571 matched payouts (including 50 of the 53 gap cases); only 3 cases (RF-1041, RF-1368, RF-1435) have true unexplained discrepancies (> INR 1). However, quoting gross figures without deduction breakdowns leads agents to perceive standard deductions as unexplained 'short payments'.

Table 1: Macro Volume vs. Operational Friction Indicators (Feb – Jun 2026)

Month (2026)	Support Intake	Finance Intake	Completed Payouts	Pending Month-End	Escalations	Escalation %
February	150	111	105	0	2	1.3%
March	146	155	148	0	23	15.8%
April	145	138	134	0	25	17.2%
May	147	124	115	0	30	20.4%
June (Spike)	**144**	**124**	**60**	**58**	**75**	**52.1%**

Key Diagnostic Takeaways

1. **Volume Invariance:** Monthly incoming refund volume is statistically constant ($\mu = 146.4$, $\sigma = 2.2$). Headcount is not undercapacity.
2. **The June Payout Collapse:** Finance payouts completed dropped by 48% in June (60 vs 115 in May), leaving 58 refunds in limbo.
3. **Escalation Composition:** Over 60% of escalations are driven solely by communication vacuums (*'chasing for update'*, *'no record in Finance'*). Eliminating information opacity resolves the majority of escalations without modifying operational staff.

2. Deep-Dive Analysis: Where the Systems Break Down

Reconciling the 755 Support rows, 689 Finance rows, and 155 Escalations with normalized identifiers reveals the precise anatomy of failure:

Table 2: Cross-Tracker Reconciliation Matrix & Failure Modes

Failure Category	Count	Root Operational Mechanism	Direct Impact on Escalations
Dropped Handoffs (Support Only)	**69** tickets	Support marks ticket 'Closed' in their sheet upon intake/forwarding; no API or unified queue pushes it to Finance.	32 escalations cite 'Ticket closed, Finance has no record' or 'Marked complete, nothing received'.
Approval Choke (Finance Pending)	**58** tickets	Approvals bottlenecked on N. Iyer (last sign-off June 13). No escalation or delegation rule exists for absences.	37 escalations cite 'Agent chasing for status update, no response' after 2+ weeks of silence.
Shadow Intake (Finance Only)	**16** tickets	Refunds triggered via off-tracker finance emails/WhatsApp. Support has zero visibility, causing confusion if queried.	Support quotes incorrect status or tells agent no request exists.
Amount Gaps & Deductions (Gross vs Net)	**53 gaps** (3 unexpl.)	53 completed payouts have gaps; deductions explain 50 of them (and 556 of 571 matched payouts overall). Only 3 are unexplained.	Escalations complain of 'Short payment' or 'Received less than promised' due to lack of deduction visibility.
Data Hygiene Deficit (Syntax & Duplication)	**22 rows** in Support	12 exact duplicates plus 10 conflicting IDs in Support (9 duplicates in Finance). Whitespace ('Closed ') and syntax drift ('RF 1079' vs 'RF-1079').	Agents get double-billed or lost in lookups; metrics are obscured.

Validation of Unlogged Shadow Cases from Brief

• **Nomad Travel (BLR-MAA):** Raised on WhatsApp June 8 (*'Bhai refund ka kya hua? BLR-MAA wala. 2 hafte ho gaye'*). Present in Escalations as **ESC-592** referencing **RF-1098**. Both *Support_Tracker* and *Finance_Tracker* have **zero record** of RF-1098. The request evaporated in WhatsApp.
• **Peak Journeys (DEL-DXB):** Emailed June 5 (*'I cancelled booking for DEL-DXB last week... no reference number'*). Logged in Escalations as **ESC-587** referencing **RF-1099**. Again, completely missing from both operational sheets.
• **Internal Support Confession (June 11):** *'Finance says they never received several refunds I closed last month'*. Confirmed quantitatively: 69 tickets marked 'Closed' by Support are absent from Finance.

What Would Be Validated With Additional Time & System Access

1. **GDS/Airline API Logs:** Determine whether airline cancellation credits were received by BharatTrip earlier, indicating Finance is sitting on settled airline funds.
2. **WhatsApp & Shared Email Inboxes:** Audit unlogged inbound messages across support channels to measure true shadow demand volume.
3. **Finance Sign-Off Authority Rules:** Clarify if sign-offs require N. Iyer due to statutory audit thresholds or if peer managers had delegation authority.

3. Target Solution: Unified Lifecycle Architecture (Zero-Headcount)

BharatTrip must transition from disjointed sheets to an **Event-Driven Single Source of Truth (SSOT)** with a deterministic state-machine and automated communication webhooks:

State-Machine: 1. INTAKE_CAPTURED (Canonical RF-ID auto-issued) → 2. ELIGIBILITY_VERIFIED (Auto-deduction calculated) → 3. FINANCE_QUEUED → 4. PAYOUT_APPROVED (Auto-batch ≤ INR 25,000) → 5. SETTLED / DECLINED (Instant agent webhook).

Table 3: Operational Trade-Off Matrix Under Zero-Headcount Constraint

Component	Implementation	Operational Rationale & Safeguard
Omnichannel Intake	**Automate (Rule-Based Parser)**	Extracts PNR, sector, agent from WhatsApp/Email, assigns canonical RF-NEW ID, auto-responds with acknowledgement. Prevents unlogged loss (RF-1098).
Deduction Engine	**Automate (Rules Engine)**	Calculates airline cancellation charges + service fee upfront. Sends itemized deduction slip, eliminating amount disputes.
Routine Approvals	**Automated Batch (≤INR 25k)**	Tickets within standard cancellation window auto-approved for payment. Removes single-manager dependency (N. Iyer bottleneck).
Disputed Refunds	**Manual (Finance Lead)**	High-value (>INR 25k) or non-refundable fare waiver requests require human scrutiny to protect cash margins.
Key Partner Relations	**Manual (Senior Support)**	High-tier travel agencies receive direct human outreach on exceptions while routine tickets flow through zero-touch pipeline.

4. Built AI Prototype: Scope & Judgement Justification

Slice Selected: An end-to-end **AI Refund Operations Copilot & Auto-Reconciler** combining: (1) A deterministic, rule-based Omnichannel Intake Agent that parses messy bilingual/Hinglish WhatsApp & email messages, extracts route sectors, scores escalation risk, issues stable unique reference IDs (RF-NEW-<HEX>), and drafts safe acknowledgement responses without premature payout commitments; (2) An automated cross-tracker reconciliation engine that flags dropped handoffs and approval stalls; and (3) An interactive Streamlit and HTML dashboard.

Judgement: Building a generic chatbot would fail because the problem is structural. This prototype attacks the two primary escalation sources: unlogged shadow requests and tracker divergence.

5. Measurement Framework & 30-Day Operational Scorecard

To verify that the process is functioning without adding staff, leadership should track two core metrics:

- **Primary Metric — Refund Escalation Rate (%):** (Monthly Escalations / Total Refund Requests) × 100.
  - *Baseline (June 2026):* **52.1%** (75 escalations on 144 requests).
  - *30-Day Target:* ≤ **8.0%** (≤12 escalations). *60-Day Target:* ≤ **3.0%**.
- **Secondary Metric — Cross-Team Handoff & Processing SLA:** Time from refund intake to payout execution.
  - *Baseline:* Average handoff lag is **2.47 days**, and total turnaround lag is **6.23 days** (with 69 tickets dropped and 58 pending payouts aging 25+ days).
  - *30-Day Target:* Handoff lag **< 15 minutes** (real-time SSOT sync); End-to-end payout turnaround ≤ **48 hours**.

Explicit Operational Assumptions

1. Standard cancellation policies carry fixed fee schedules allowing automated deduction calculation for >80% of routine domestic bookings.
2. Travel agents accept WhatsApp/Email automated confirmations provided a valid tracking ID and clear acknowledgment are supplied.
3. Finance can execute batch payouts based on system-validated rules without manual per-item executive re-authorization.